Advanced Prompt Engineering & Reasoning Handbook

Few-Shot, Chain-of-Thought (CoT), System Framing & Guardrails

SkillForge AI Engineering Series - Certified Curriculum Materials

1. The Persona & Context Framing Pattern

System Message: Establish authority, strict output constraints, domain scope, and tone.
Delimiters: Use triple quotes or XML tags <context>, <input> to prevent prompt injection.
Negative Constraints: State what NOT to do explicitly to guide probability distributions.

2. Chain-of-Thought (CoT) & Structured Reasoning

Zero-Shot CoT: Append Let us think step by step to trigger multi-hop intermediate deductions.
Few-Shot Exemplars: Provide 3-5 verified input-reasoning-output triplets for strict pattern following.
Self-Consistency: Sample multiple reasoning paths at temperature 0.7 and take majority consensus.

3. Structured JSON Outputs & Function Calling

Enforce strict JSON schemas using Pydantic / Zod for deterministic API interop.
Tools and Function Definitions: Provide accurate descriptions so the model can invoke tools.